

Transform Calculus

(MA20202)

Assignment-2

1. Show that

$$1 * 1 * \cdots * 1 (n\text{-times}) = \frac{t^{n-1}}{(n-1)!}.$$

2. Using the convolution theorem, show that

$$\int_0^t \sin u \cos(t-u) du = \frac{1}{2} t \sin t.$$

3. Find the inverse Laplace transform of $F(s) =$

i) $\frac{s-1}{(s+3)(s^2+2s+2)},$

ii) $\frac{se^{-2s}}{s^2+3s+2},$

iii) $\frac{2s+6}{(s^2+6s+10)^2},$

iv) $\ln\left(\frac{s+2}{s+1}\right),$

v) $\frac{1}{(s+3)(s-1)},$

vi) $\frac{s}{(s^2-a^2)^2}.$

4. i) If the Laplace transform of $f(t)$ is $F(s)$, then prove that the inverse Laplace transform of $\frac{F(s)}{s^2}$ is

$$\int_0^t \left\{ \int_0^{t_1} f(\tau) d\tau \right\} dt_1.$$

- ii) If the Laplace transform of $f(t)$ is $F(s)$, then show that the Laplace transform of $t^2 f''(t)$ is $s^2 F''(s) + 4s F'(s) + 2F(s)$.

5. i) Find the Laplace transform of the Bessel function $\mathcal{J}_0(t)$ of order 0, which is defined by the following infinite series

$$\mathcal{J}_0(t) = 1 - \frac{t^2}{2^2} + \frac{t^4}{2^2 4^2} - \frac{t^6}{2^2 4^2 6^2} + \cdots$$

Then find the Laplace transform of $\mathcal{J}_0(at)$.

- ii) Find the Laplace transform of the function $\sin \sqrt{t}$.

6. Evaluate the following integrals using the Laplace transform method:-

i) $\int_0^\infty t e^{-2t} \cos t dt$

ii) $\int_0^\infty \frac{e^{-3t} - e^{-6t}}{t} dt$

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